A proposal for the AI Project

"ActSmart: Asr, LLM and TTS for human machine interaction"

Submitted by

Keval DineshBhai Viradiya
Devanshi VijayKumar Trivedi

Supervised by

Prof. Dr.-Ing. Christian Bergler

Ostbayerische Technische Hochschule Amberg-Weiden
Department of Electrical Engineering, Media and Computer Science

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Abstract

The ActSmart project aims to create a smart healthcare assistant that makes it easier for patients to manage their health. By using **Machine Learning (ML)** technologies such as **Natural Language Processing (NLP)** and **Automated Speech Recognition (ASR)**, ActSmart offers features such as automated consultations, symptom checking, and health education. It relies on advanced AI models like OpenAI Whisper, Meta Llama, and other deep learning tools to provide smooth, user-friendly interactions and accurate health advice.

The goal of ActSmart is to connect between healthcare and technology, making it simple and efficient for people to access important health information and guidance when they need it.Combining ASR, NLP, and Text-to-Speech (TTS) technologies, ActSmart takes a comprehensive and integrated approach to healthcare management. It empowers users by providing accurate symptom evaluations, personalized advice, and practical insights through smooth voice-based interaction. ActSmart isn't just a tool for basic health advice—it's a step toward promoting self-awareness, preventive care, and tailored health management. Built with Python and open-source tools, the project seeks to transform the way people interact with technology in the healthcare space, making health guidance accessible, reliable, and easy to understand for everyone.

1 Introduction and Motivation

Human-machine interaction is becoming more important in today's technology, with examples like virtual assistants and automated customer support. However, these systems are often limited to basic tasks like automating actions or retrieving information. Creating a system that can truly understand natural language, grasp context, and respond in a natural, human-like way is still a big challenge. It involves combining three advanced technologies: speech recognition, understanding context, and text-to-speech conversion.
ActSmart tackles these challenges by combining advanced technologies like **Automatic Speech Recognition (ASR)**[1], **Large Language Models (LLMs)** [2], and **Text-to-Speech (TTS)**[3] into a Python-based platform. Itis designed to create smooth, natural, and user-friendly interactions between people and machines, with a special focus on healthcare. Since healthcare often requires quick, accurate, and easy-to-access information something traditional systems struggle to provide, ActSmart simplifies health-related interactions to make vital information more accessible and useful for everyone.
ActSmart is designed to make healthcare advice more accessible and personalized. With healthcare systems under pressure, people are increasingly turning to technology for basic health consultations and guidance.ActSmart is designed to provide easy to access and understand personalized health advice .With healthcare systems under pressure, ActSmart provides personalized guidance tailored to your specific health concerns. It uses advanced tools like OpenAI Whisper to understand your speech and Meta Llama to grasp the context of your situation. This makes ActSmart more advanced than what's currently available, aiming to change the way we interact with technology in healthcare and setting a new standard for smarter, more intuitive health support

2 Related Work

There are a lot of apps these days that use smart technologies to help with health-related issues. For example, **IBM Watson Health** helps doctors by analyzing tons of medical data to support their decision-making, while **Ada Health**[4] allowed the user to determines their symptoms with an easy-to-use interface.

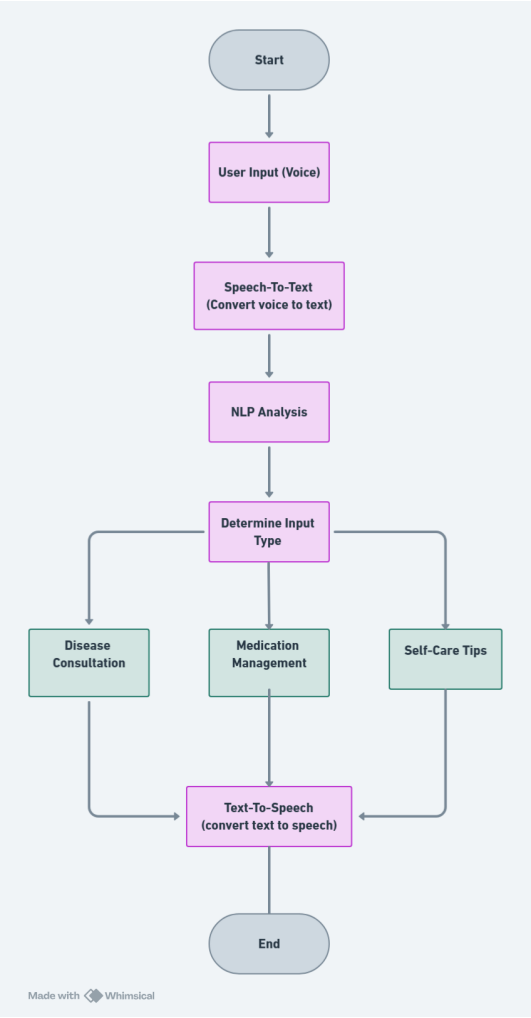
What sets SmartAct apart is its ability to perform a range of functions, such as predicting diseases and assessing symptoms, by combining Automatic Speech Recognition (ASR), NLP, and Text-to-Speech (TTS) technologies. Unlike other apps that focus on specific health tasks, SmartAct brings everything together into one platform, making it simpler for users to manage their health. The system listens to what the user says, picks up on the symptoms, guesses what might be wrong, then gives advice that's specific to them. It also speaks that

advice back in a clear way so it is easy to understand. By using smart tools such as Meta-Llama[5], Whisper[6], and gTTS[7], SmartAct ensures everything is both accurate and easy to use.

By combining these advanced technologies and addressing multiple health-related needs in one application, SmartAct offers a more holistic and personalized healthcare management solution compared to existing applications.

3 Methodology

SmartAct's development follows a well-thought-out process to create a smart healthcare assistant that helps users manage their health. This process can be broken down into several key steps, each designed to ensure the system is both effective and user-friendly.



3.1 Processing:

The data collection process starts by capturing user symptoms through speech. OpenAI's Whisper technology is used for speech recognition, ensuring accurate conversion of speech to text. After that, the text is cleaned up in a preprocessing step, where unnecessary words are removed so the system can focus only on the relevant symptoms. This makes sure that only the important details are used for further analysis.

3.2 Symptom Analysis:

The cleaned symptoms are matched against a pre-existing dataset that maps symptoms to their associated diseases[8]. This matching process leverages the dataset to identify potential connections between the user-provided symptoms and known medical conditions. To determine the most relevant conditions, the system uses a Counter to analyze the frequency of occurrences of conditions associated with the input symptoms. Additionally, a minimum match threshold is implemented to ensure that only the most relevant and significant conditions are considered, enhancing the accuracy and reliability of the identification process.

3.3 Health Advice Generation:

The cleaned symptoms are compared to a pre-existing dataset that links symptoms to diseases. This matching process helps identify possible connections between the user's symptoms and known conditions. To find the most relevant conditions, the system looks at how often certain conditions appear with those symptoms. A minimum match threshold is also used to ensure that only the most significant conditions are considered, improving the accuracy and reliability of the results.

3.4 Text-to-Speech Conversion:

The system transforms the health advice into speech using Google Text-to-Speech (gTTS), so that it's accessible to everyone, especially people with visual impairments or those who simply prefer hearing the information. Once spoken, the advice is saved as an audio file, making it easy for users to play back whenever they need it. This feature not only simplifies how the information is delivered, but also ensures that it's easy for anyone to access and understand the health guidance.

3.5 Speech Recognition:

When a user speaks their symptoms, the system uses advanced speech recognition to accurately transcribe what's said. This technology turns the spoken

words into text, allowing the system to understand and analyze the information. Accurately capturing the symptoms is a vital first step, ensuring the system can identify possible health conditions and offer helpful advice tailored to the user's needs.

4 Experiments

To evaluate the effectiveness and reliability of SmartAct, several experiments will be conducted, focusing on different aspects of the system.

Accuracy of Speech-to-Text: The first test will evaluate how well the system transcribes spoken language into text, particularly when handling various accents and audio quality levels. We'll use a variety of audio samples, including recordings with background noise and unique accents, for this evaluation. The performance of the speech recognition system, powered by OpenAI's Whisper, will be measured using two main metrics: word error rate (WER) and overall accuracy. This test is crucial to ensure that SmartAct can understand and process speech input in different situations and from diverse users..

Health Condition Prediction: The next test will check how well the system can predict health conditions based on symptoms reported by users. To do this, we'll compare its predictions with a trusted dataset of symptoms and conditions. We'll use metrics like precision and recall to measure how accurate these predictions are. Tools like pandas and scikit-learn will help with the analysis, and the Meta-Llama-3-8B-Instruct model will be used to provide health advice. This test will ensure that SmartAct gives reliable and accurate health assessments.

User Testing for Voice Quality: We'll ask users to share their thoughts on the voice quality of the health advice. They'll listen to the advice and rate how clear, natural, and pleasant it sounds in a survey. The feedback will help us improve the voice to make it more user-friendly. We'll use Google Text-to-Speech (gTTS) to generate the voice, and the survey results will show us where we can make improvements. This test is all about making the listening experience better for users.

System Speed: Lastly,We will also test how fast the system responds to ensure it works smoothly in real-time. This includes measuring how quickly it performs tasks like converting speech to text, analyzing symptoms, and generating advice. We'll use Python to create tools that monitor performance. This test will make sure SmartAct handles user interactions quickly without noticeable delays.

4.1 Tools and Libraries Used:

Programming Language: Python

Libraries: pandas, collections, gTTS, os, transformers, torch, speech recognition

Models: Meta-Llama-3-8B-Instruct, OpenAI Whisper,spaCy

These tests will help SmartAct demonstrate that it can accurately turn speech into text, predict health conditions, provide clear voice responses, and react quickly to user input. By going through this testing, we'll make sure the system works smoothly, is reliable, and easy for everyone to use.

5 Result and Discussion

5.1 System Functionality

The system brings together different features to provide a smooth health advisory experience. It turns spoken audio into text, analyzes symptoms, and identifies possible health conditions based on the data. It then generates personalized health advice using a pre-trained LLaMA model and converts that advice into audio using gTTS.

5.2 Key Achievements

The system brings together several components to provide a smooth health advisory experience. It uses the SpeechRecognition library to accurately convert audio inputs into text, which is then analyzed. During preprocessing, irrelevant words are filtered out to ensure that the relevant symptoms match the dataset for precise condition identification. The system looks at the symptoms and suggests possible health conditions, giving priority to the most likely ones. After that, it provides advice, offering things like diet tips, over-the-counter medications, self-care ideas, and guidance on when to see a doctor. Lastly, it uses gTTS to turn the advice into speech, making it easier to use, particularly for people with visual impairments.

5.3 Output

Recognized Text: I have a cold fever and body pain but not that much headache
Possible conditions: acute bronchitis, anxiety, strep throat Setting pad token id to eos token id:None for open-end generation.
Generated Advice:
Food Recommendations:
1. Stay hydrated: Drink plenty of fluids, such as water, clear broth, or electrolyte-rich beverages like coconut water or sports drinks. Aim for at least 8-10 glasses of fluid per day.
2. Ease digestion: Opt for bland, easily digestible foods like bananas, rice, applesauce, and toast (BRAT diet). Avoid spicy, fatty, or heavy foods that can

exacerbate symptoms.
3. Boost immunity: Incorporate immune-boosting foods like citrus fruits (e.g., oranges, grapefruits), leafy greens (e.g., spinach, kale), and fermented foods (e.g., yogurt, kefir) into your diet. **Medicines:*
1. Pain relief: Over-the-counter pain relievers like acetaminophen (Tylenol) or ibuprofen (Advil, Motrin) can help alleviate headaches and body aches. Always follow the recommended dosage and consult your doctor if necessary.
2. Fever reduction: If your fever is above 102°F (39°C), you can take over-the-counter medications like acetaminophen or ibuprofen. However, always consult your doctor before giving medication to children or if you have a history of kidney or liver disease. **Self-Care:*
1. Rest: Get plenty of rest to help your body recover. Aim for 8-10 hours of sleep per night.
2. Hygiene: Practice good hygiene by washing your hands frequently, especially after blowing your nose, coughing, or sneezing.
3. Stress management: Engage in stress-reducing activities like meditation, deep breathing, or yoga to help manage anxiety and promote relaxation.
4. Monitor symptoms: Keep track of your symptoms and consult your doctor if they worsen or persist. If you experience difficulty breathing, chest pain, or severe headache, seek immediate medical attention.

Remember to consult your doctor if your symptoms persist or worsen over time. They can provide personalized guidance and help you determine the best course Please consult a healthcare professional for specific advice based on your symptoms.

6 Summary, Future Work and Conclusion

6.1 Summary:

The system uses different technologies to give health advice based on the symptoms the user shares. It turns speech into text, processes the symptoms, identifies possible conditions, gives health tips, and then reads the advice out loud. Speech Recognition makes sure the user's words are transcribed correctly, and the system filters out extra words to focus on the key symptoms. The system uses a pre-existing dataset to identify possible health conditions and then creates personalized advice, including self-care tips, food suggestions, and when to seek medical help. The gTTS module makes the system more accessible by turning the advice into audio, which is helpful for users with visual impairments or reading difficulties. Overall, the project shows that AI can be used to provide basic health guidance effectively.

6.2 Future Work:

The system does a good job right now, but there's a lot of room for improvement. For example, adding more symptoms and conditions to the dataset would make the system even better. It would also be useful to improve the speech-to-text feature so it can handle different languages and accents, making it easier for more people to use. To make the system better, we could offer advice based on things like medical history and allergies. It would also help if the system gave faster responses. Lastly, letting users share their feedback would help improve the system over time.

6.3 Conclusion:

This project shows how artificial intelligence and natural language processing can work together to give basic health advice. By analyzing symptoms and generating health tips automatically, the system provides an easy and accessible way for people to get medical insights. Using tools like Speech Recognition, LLaMA, and gTTS makes the system more useful. While it's a good start, more improvements are needed to overcome its current limits and make it more useful. With future updates, this system could become an important tool in personalized healthcare.

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